Lecture #15

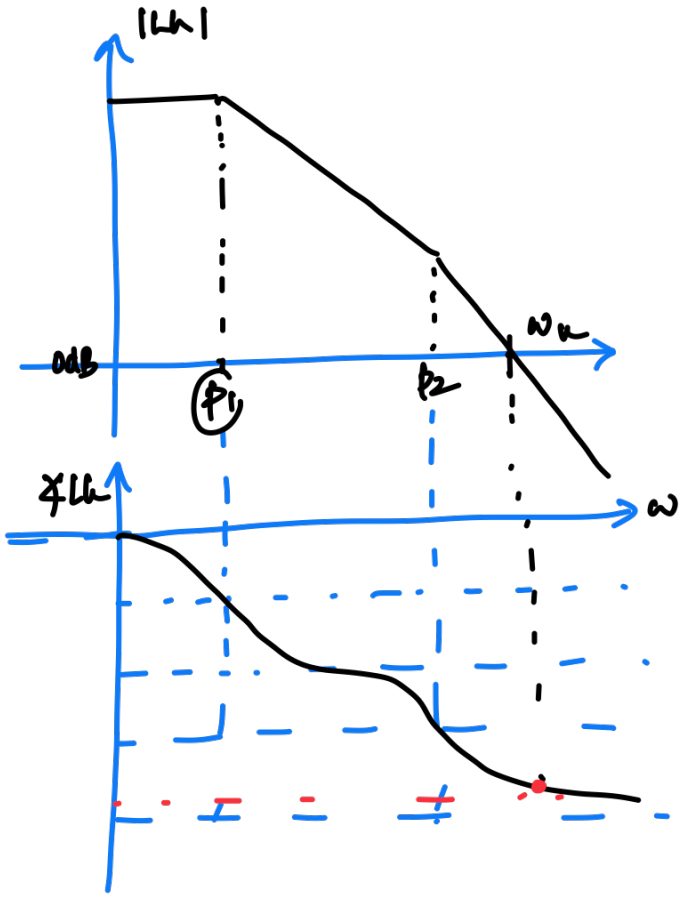
$$H(s) = \frac{L_u}{1+L_u}$$

$$D(s) = 1 + L_u = 0$$

$$L_u = \frac{A_0/\tau}{\left(1+\frac{s}{p_1}\right)\left(1+\frac{s}{p_2}\right)}$$

$$|L_u(j\omega_u)| = 1$$

$$\angle L_u(j\omega_u)$$



$$H(s) = \frac{L_u}{1+L_u}$$

$$D(s) = 1 + \frac{A_0}{\tau}\left(1+\frac{s}{p_1}\right)\left(1+\frac{s}{p_2}\right) = 0$$

$$\frac{A_0}{\tau}\left(1+\frac{s}{p_1}+\frac{s}{p_2}+\frac{s^2}{p_1 p_2}\right)$$

$$= 1+\frac{s}{p_1}+\frac{s}{p_2}+\frac{s^2}{p_1 p_2}+\frac{A_0}{\tau}$$

$$ax^2+bx+c=0$$

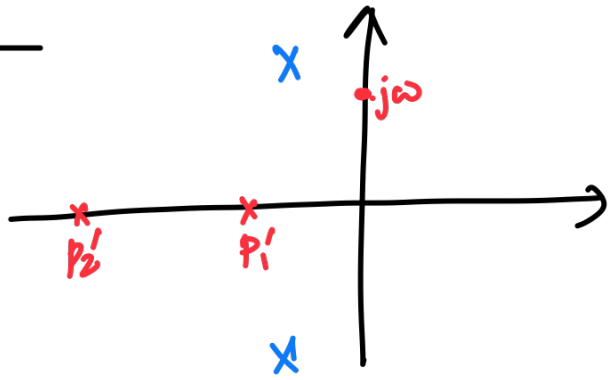
$$x_1+x_2 = -\frac{b}{a} \xrightarrow{x_2 \gg x_1} x_1+x_2 \approx x_2 = -\frac{b}{a}$$

$$x_1 x_2 = \frac{c}{a} \longrightarrow x_1 = -\frac{c}{b}$$

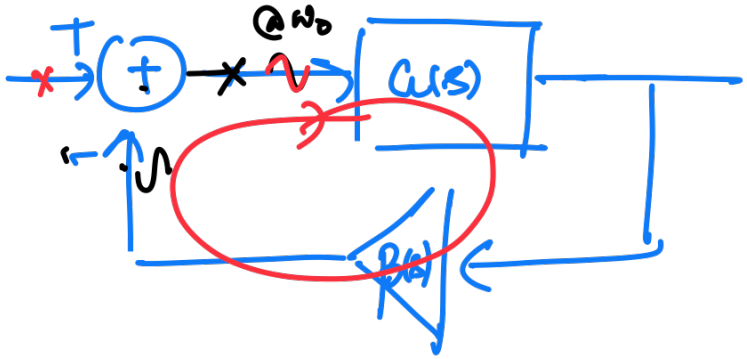
$$p_2' = -\frac{\left(\frac{1}{p_1}+\frac{1}{p_2}\right)}{\frac{1}{p_1 p_2}} = -(p_1+p_2) \approx -p_2$$

$$p_1' = -\frac{\left(1+\frac{A_0}{\tau}\right)}{\left(\frac{1}{p_1}+\frac{1}{p_2}\right)} \approx -\frac{A_0/\tau}{\frac{1}{p_1}} = -\frac{A_0 p_1}{\tau}$$

$$H(s) = \frac{}{(s-p_1')(s-p_2')}$$

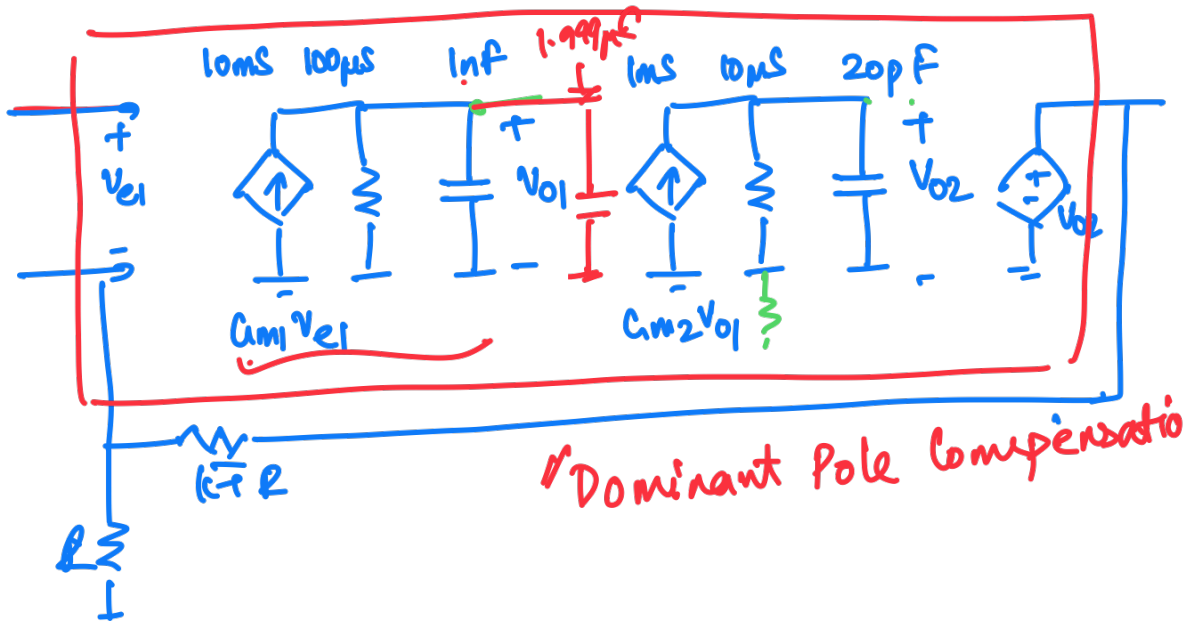


$$H(jp_1') = \frac{}{(jp_1'-p_1')(jp_1'-p_2')}$$



$|L_H(j\omega_0)| = 1$ & $\angle L_H(j\omega_0) = (2k+1)\pi$ ✓

"Barkhausen Criterion"



"Dominant Pole Compensation"

$k = 4$

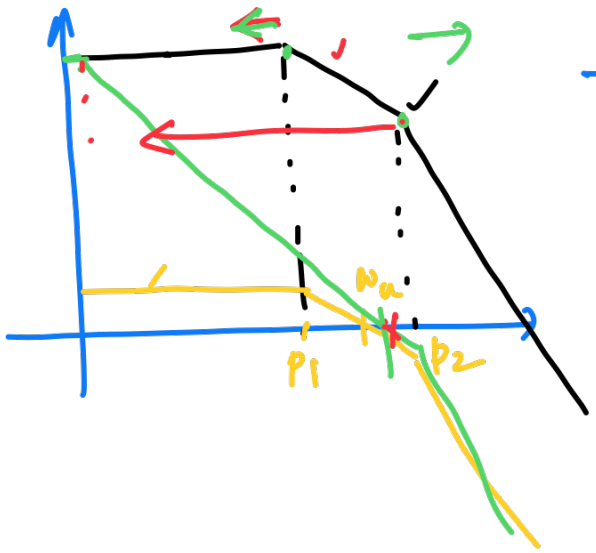
$$A(s) = \frac{G_{m1}r_{o1}}{1 + s r_{o1}C_{o1}} \times \frac{G_{m2}r_{o2}}{1 + s r_{o2}C_{o2}} = \frac{10^4}{(1 + s/p_1)(1 + s/p_2)}$$

$s \to j\omega$

$$G_{m1}r_{o1} = \frac{10\,mS}{100\,\mu S} = 100 \quad ; \quad G_{m2}r_{o2} = \frac{1mS}{10\mu S} = 100$$

$$p_1 = \frac{1}{r_{o1}C_{o1}} = \frac{100 \times 10^{-6}}{1 \times 10^{-9}} = 10^5 \text{ rad/s}$$

$$p_2 = \frac{1}{r_{o2}C_{o2}} = \frac{10 \times 10^{-6}}{20 \times 10^{-12}} = \frac{10^6}{2} = 5 \times 10^5 \text{ rad/s}.$$



- if $p_1 \le \omega_u < p_2$

$$LG = \frac{10^4/4}{(1 + s/p_1)(1 + s/p_2)}$$

$$|LG(j\omega_u)| = 1$$

$$|LG(j\omega_u)| = \frac{10^4/4}{\left|\left(1 + j\frac{\omega_u}{p_1}\right)\right| \left|\left(1 + j\frac{\omega_u}{p_2}\right)\right|}$$

$\frac{\omega_u}{p_1} \gg 1$; $\frac{\omega_u}{p_2} \ll 1$

$$\approx \frac{10^4/4}{\frac{\omega_u}{p_1}} = 1$$

$$\Rightarrow \omega_u = 2500 \times p_1$$

$\omega_u = 2500 \times 10^5$ rad/s. $> p_2$ ✗

— if $\omega_u > p_2 > p_1$

$$|L_h(j\omega_u)| = \frac{10^4/4}{\left(1+j\frac{\omega_u}{p_1}\right)\left|1+j\frac{\omega_u}{p_2}\right|}$$

$$\approx \frac{10^4/4}{\frac{\omega_u}{p_1} \times \frac{\omega_u}{p_2}} = 1$$

$$\omega_u = \sqrt{2500\, p_1 p_2} = \sqrt{2500 \times 10^5 \times 5 \times 10^5}$$

$$= 50 \times 10^5 \times 2.236$$

$$= \underline{112 \times 10^5} \text{ rad/s.} \quad > p_2 > p_1$$

$$\angle L_h(j\omega_u) = -\tan^{-1}\left(\frac{\omega_u}{p_1}\right) - \tan^{-1}\left(\frac{\omega_u}{p_2}\right)$$

$$= -\tan^{-1}(112) - \tan^{-1}\left(\frac{112}{5}\right)$$

$$= -90^\circ - 87^\circ = -177^\circ$$

$$\phi_m = \angle L_h(j\omega_u) - (-180^\circ) = \underline{3^\circ}$$

— Make one pole dominant.

$$L_h = \frac{A_{DC}}{\left(1+\frac{s}{p_1'}\right)\left(1+s/p_2'\right)} \quad ; p_2' = p_2$$

$\phi_m = 76^\circ$ $\quad p_1' < \omega_u' < p_2'$

$$|L_a(j\omega_n')| = 1$$

$$\approx \frac{10^4/4}{\frac{\omega_n'}{p_1'} \times 1} = 1$$

$$\omega_n' = 2500\, p_1'$$

$$\phi_m = \angle L_a(j\omega_n') - (-180^\circ)$$

$$= 180^\circ - \tan^{-1}\left(\frac{\omega_n'}{p_1'}\right) - \tan^{-1}\left(\frac{\omega_n'}{p_2'}\right)$$

$$= 180^\circ - \tan^{-1}(2500) - \tan^{-1}\left(\frac{2500\, p_1'}{p_2'}\right)$$

$$78^\circ = 90^\circ - \tan^{-1}\left(\frac{2500\, p_1'}{p_2'}\right) \quad = \tan^{-1}\left(\frac{p_2'}{2500 p_1'}\right)$$

$$\tan^{-1}\left(\frac{2500\, p_1'}{p_2'}\right) = 14^\circ$$

$$\phi_m = 90^\circ - \tan^{-1}\left(\frac{\frac{A_0}{K} p_1'}{p_2'}\right)$$

$$\Rightarrow \tan^{-1}\left(\frac{p_2'}{\frac{A_0}{K} p_1'}\right) = \phi_m$$

$$\frac{p_2'}{2500\, p_1'} = 4 \Rightarrow p_1' = \frac{5\times 10}{2500 \times 4}$$

$$= 50 \text{ rad/s} = \frac{1}{r_{01} C_{01}'}$$

$$50 = \frac{100 \times 10^{-6}}{C_{01}'}$$

$$C_{01}' = 2 \times 10^{-6} = 2\mu f.$$

$$\frac{p_2'}{\frac{A_0}{K} p_1'} = 4$$

$$\frac{p_2'}{\omega_n'} = \frac{p_2'}{\frac{A_0}{K} p_1'} = \tan(\phi_m)$$

ϕ_m	p_2'/ω_n'
45°	1
30°	$1/\sqrt{3}$
60°	$\sqrt{3}$
78°	4

$C = 2\mu F$

C per unit area = $2 ff / \mu m^2$

$$\frac{2\mu F}{2 \times fF} \times \mu m^2 = 10^9 \, \mu m^2$$

$3.13 \times 10^4 \, \mu m \times 3.13 \times 10^4 \, \mu m$